

EP 502: Informatics for Materials Design

Course Code	EP 502
Course Name	Informatics for Materials Design
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	BTech 4th Year, M.Sc. Physics, PhD Scholars, M.Tech
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Computational material science:** Crystal Structure and symmetry, Material properties, Property based classification of materials (mechanical, electrical, thermal, magnetic, optical), Performance of materials, Meta materials, Need for new materials. (4 Hours)
- **State of art techniques at different length scales:** Concept of multiscale modeling, First principles approach, Density Functional Theory (electronic level), Brief introduction to Schrodinger's equation, Overview of most commonly used approximations (Born Oppenheimer, Local Density Approximations), Kohn-Sham equations, Pseudopotentials, Description of the self-consistent field iterations, Total energy minimization, Overview of major algorithms in DFT calculations. (9 Hours)
- **Databases and Python Scripting:** DBMS fundamentals, Design, Workflows, Query writing, python libraries: Numpy, Panda, Pymatgen, Materials database repositories, Materials open database integration APIs. (6 Hours)
- **Introduction to Machine learning for material design:** Philosophy behind machine learning, Basic vocabulary terms, Algorithms based on learning: supervised and unsupervised, Regression vs. classification, Regression algorithms, Clustering algorithms, Decision tree algorithms, Interpretability analysis using Lyme/Shap. Model independent Descriptors for material data analytics. (9 Hours)

Laboratory Modules

1. Lab work (1 Credit)
2. Hands on with Quantum Espresso (QE)- 3 Labs
3. Hands on with MySQL- 1 Lab
4. Working with python scripts, use of APIs etc – 2 Labs
5. Creating databases using APIs to fetch material data – 1 Lab
6. Machine learning with Scikit/Weka – 2 Labs

Research project (1 Credit)

Based on use of machine learning/Quantum Espresso for understanding material design and its properties for particular applications like magnetic storage, photovoltaic response, electrical conductivity, magnetism and spintronic application.

Text books:(Relevant and Latest, Only 2)

1. June Gunn Lee, **Computational Materials science**, CRC press, 2012.
2. Aurélien Géron, **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow**, 2nd Edition, O'Reilly Media, Inc. 2019

References

1. Online resources for learning SQL, python
2. Research papers